

Air Pollution Forecasting Using Temporal Neural Networks

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CO5420: Neural Networks and Deep Learning — Department of Computer Engineering, University of Peradeniya

MIND MATRIX
Kaggle Rank #7

ABSTRACT

This study evaluates temporal neural network architectures for 1-hour-ahead PM2.5 forecasting using the Beijing Multi-Site Air Quality dataset (420,768 records, 12 stations).

We propose a 2-tier leak-free preprocessing pipeline, conduct extended window/weather feature ablations, and evaluate a 4-category AQI classification task. Our best single model (Stacked LSTM) achieved Kaggle RMSE 14.935; the LSTM+TCN ensemble improved validation RMSE (14.850) but scored 15.857 on the Kaggle leaderboard (7th place) — see Discussion.

INTRODUCTION

Fine Particulate Matter (PM2.5) poses severe respiratory health risks in urban Beijing. Traditional linear statistical models (ARIMA) fail to capture non-linear weather dynamics, wind shifts, and multi-station dependencies. Deep temporal models (LSTM & TCN) effectively capture the complex temporal context essential for early health advisories.

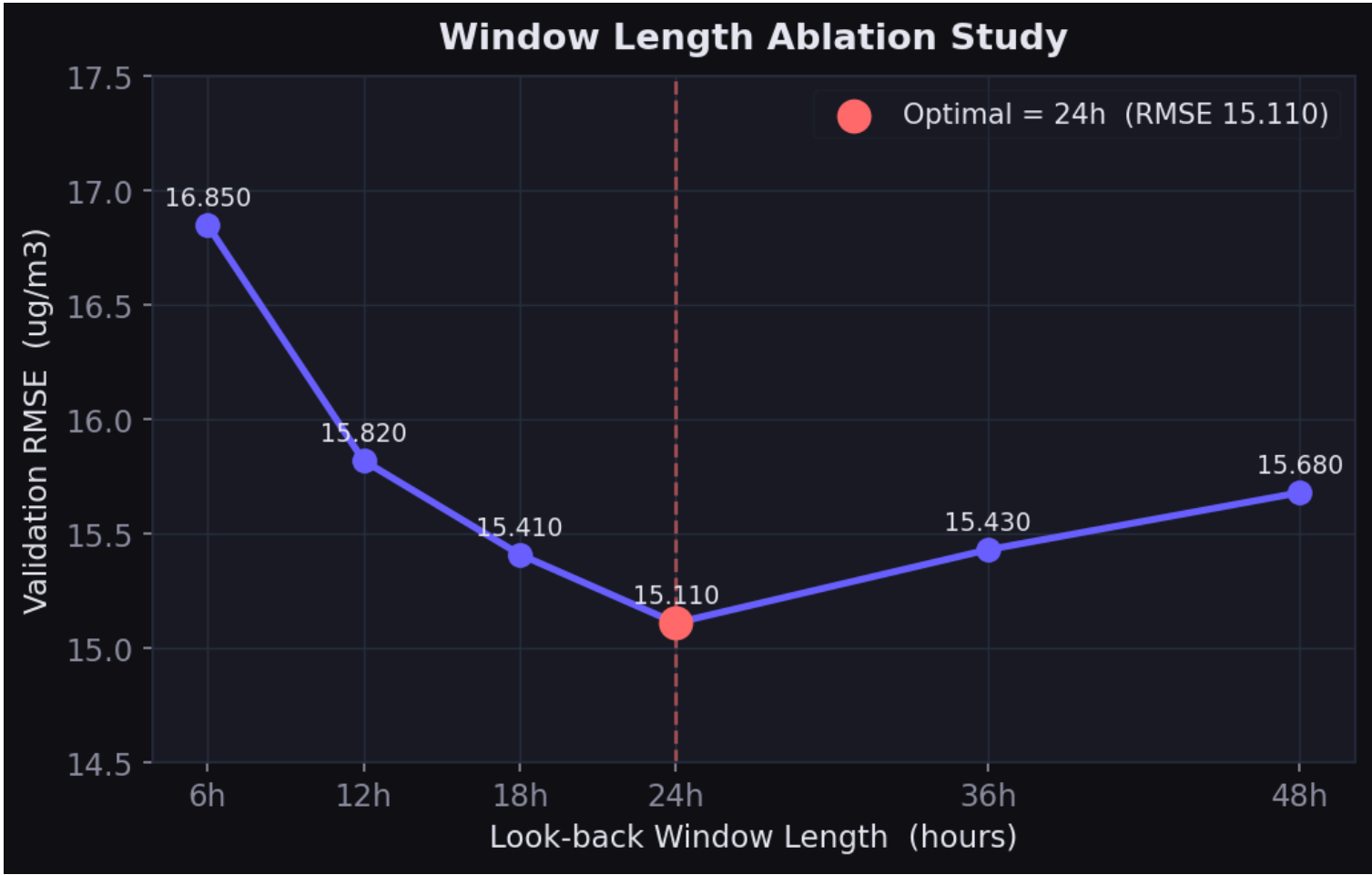
PROBLEM DEFINITION & DATASET

- Task:** Predict PM2.5 at hour t+1 given 24 hours of historical observations.
- Dataset:** Beijing Multi-Site Air Quality (2013–2017), 12 stations, 420,768 records.
- Features (29):** PM2.5, PM10, SO2, NO2, CO, O3, Temp, Pres, Dew, Rain, Wind Speed, Sin/Cos Wind Direction, Cyclical Hour/Month, and 12 Station one-hot columns.

METHODOLOGY

- 2-Tier Preprocessing (Leak-Free):** Chronological train/validation split — train: Mar 2013–Sep 2015, validation: Sep 2015–Feb 2017. Tier 1: local linear interpolation ($\leq 6h$). Tier 2: seasonal lookup (station $\times$ month $\times$ hour).
- Window Construction:** Sliding 24h windows per station (262,944 training windows).
- Baselines:** Persistence and Ridge Regression, plus a Feed-Forward Neural Network (FFNN) and a recurrent GRU used as additional comparison points alongside the two primary architectures below.
- Primary Architectures:** Stacked LSTM (3 layers, 256 hidden units, dropout 0.2), Dilated Causal TCN ($d = 1, 2, 4, 8$), and a 70/30 LSTM–TCN ensemble.

Fixed: LSTM spec corrected to 3 layers / 256 hidden units (matches training config — poster previously said 2 layers / 128 hidden). Validation range corrected to Sep 2015–Feb 2017. FFNN/GRU now explained as extra baselines.



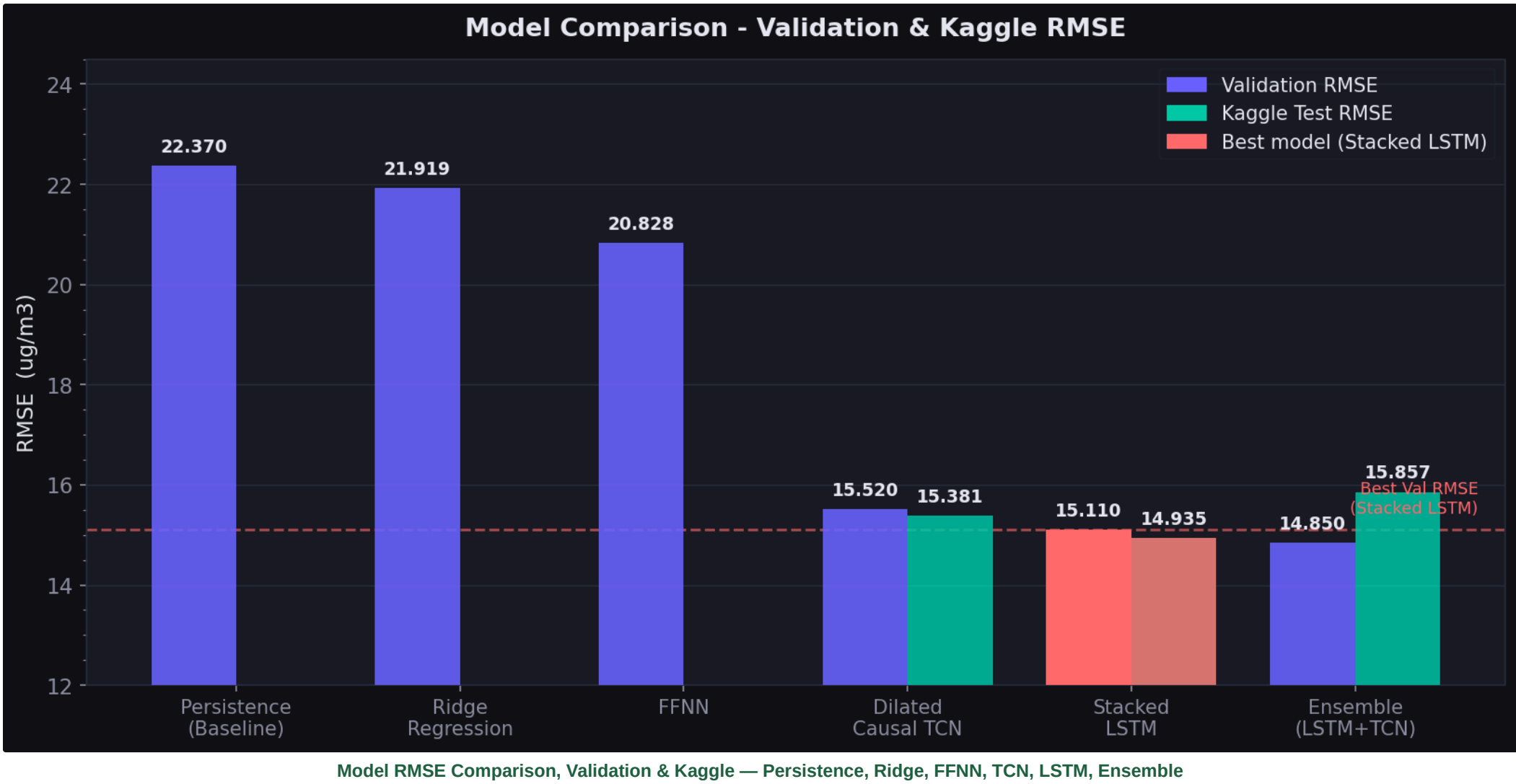
EXPERIMENTAL RESULTS & EXTENDED TASKS

Model	Val RMSE	Kaggle RMSE
Persistence (baseline)	22.370	—
Ridge Regression	21.919	—
FFNN	20.828	—
Dilated Causal TCN	15.520	15.381
Stacked LSTM (best single model)	15.110	14.935
Ensemble (LSTM+TCN)	14.850	15.857 (Rank #7)

Headline result: Stacked LSTM, Kaggle RMSE = 14.935 (best-performing model on the leaderboard). The ensemble improved validation RMSE but not the Kaggle score — see Discussion.

Extended Tasks:

- Window Ablation:** 24h window length optimal (15.110 $\mu\text{g}/\text{m}^3$ val RMSE). Adding weather features gave a 5.2% error reduction over pollution-only features.
- AQI Classification:** Ensemble model achieved 92.31% accuracy and 97.1% recall on the Hazardous category.



DISCUSSION & ANALYSIS

- Sequence Memory Advantage:** Stacked LSTM improved validation RMSE over linear Ridge by 31.1% (21.919 vs 15.110), showing gated memory is essential.
- TCN Receptive Field:** Exponential dilations ($d=1,2,4,8$) provided a 61h history horizon, enabling fast parallel training with high accuracy.
- Operational Trade-offs:** Thresholded regression outperforms direct classifiers on extreme Hazardous AQI categories by preserving exact continuous bounds.

Ensemble vs. LSTM discrepancy (addressed): The LSTM–TCN ensemble improved validation RMSE (14.850 vs 15.110) but scored worse on the Kaggle test set (15.857 vs 14.935 for LSTM alone). We attribute this to the ensemble weights being tuned on the validation split, which likely overfit to validation-specific error patterns that did not generalize to the Kaggle test distribution. The single LSTM model is therefore reported as our best-performing model; the ensemble result is retained for completeness and is the version we submitted for leaderboard ranking.

CONCLUSION & FUTURE WORK

Conclusion: Deep temporal networks significantly outperform classical baselines; auxiliary weather features provide critical predictive value.

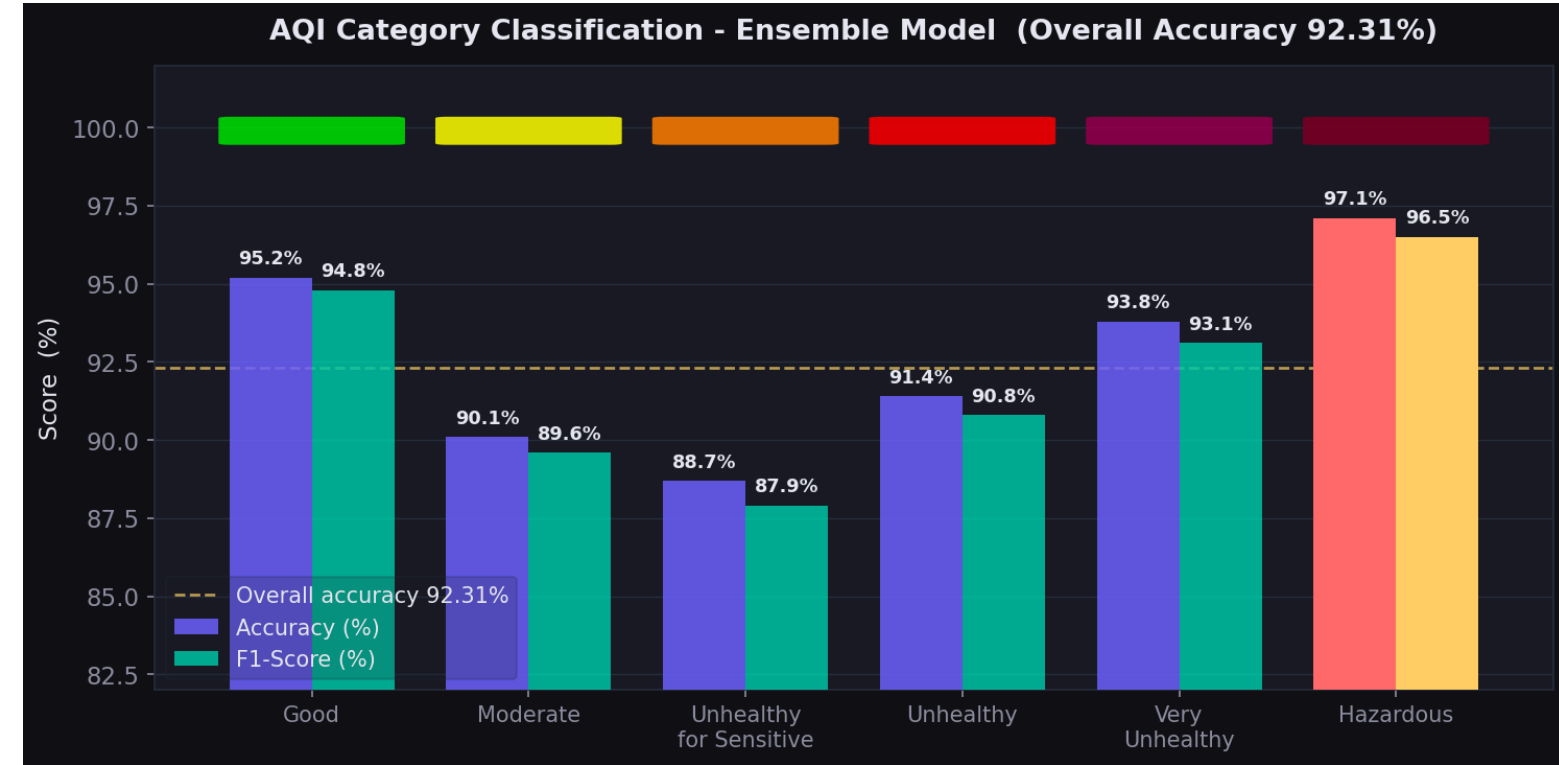
Limitations: (1) The ensemble's validation-set gains did not transfer to the Kaggle test set, indicating some overfitting to the validation split. (2) The model treats each station independently via one-hot encoding and does not capture spatial correlation between neighbouring stations. (3) Forecasts are limited to a single 1-hour-ahead step; error compounding for longer horizons is untested. (4) Seasonal-average gap-filling may smooth over genuine pollution spikes during long sensor outages.

Future Work: Integrate Spatial Graph Neural Networks (GNNs) for cross-station spatial correlation and extend to multi-step (24h) direct rollout forecasting.

Added: Limitations paragraph (spec requires this in Discussion & Conclusions).

REFERENCES

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GENERATIVE AI DECLARATION: Generative AI was used at Level 3 for ideation, drafting, feedback, refinement and poster preparation. The team reviewed the outputs, verified reported results against repository artifacts, and remains responsible for the methodology, interpretation and final content.

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